# GOLDBOD TAS

## Criterion III.3 Response

Performance Validation: Speed, Response Times, Scalability, Stability and Compatibility

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GR/GGB/TS/NCT/042/2026 | April 2026

| III.3 | Criterion<br>**Performance Validation: Speed, Response Times, Scalability, Stability and Compatibility** |
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## Deployment Environment

The system is currently deployed on a Virtual Private Server with Nginx as the reverse proxy, PM2 for Node.js process management, and PostgreSQL as the primary data store. TLS is provisioned via Certbot. The architecture is explicitly designed for cloud migration to AWS Africa (af-south-1) for national go-live, where all SLO targets are met or exceeded under managed infrastructure.

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| **Web server** | Nginx — reverse proxy, static asset serving, SSL/TLS 1.3 termination, gzip compression enabled |
| **Process manager** | PM2 — Node.js process clustering, automatic restart on failure, memory threshold monitoring |
| **Database** | PostgreSQL 15 — connection pooling via pg-pool, indexed on actor_id, licence_number, trace_id, created_at |
| **API runtime** | Node.js 20 LTS / TypeScript — stateless request handling, no shared mutable state, horizontally scalable |
| **Frontend delivery** | React.js compiled to static assets — CDN-cacheable, lazy-loaded route bundles, initial payload < 200 KB |
| **TLS** | TLS 1.3 enforced on all endpoints via Certbot / Let's Encrypt — A-grade SSL rating |

## Performance Benchmarks

The following results were measured against the live deployed instance. API endpoints tested using concurrent request simulation:

| Metric | Result |
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| **Dashboard load (cold)** | < 1.8 seconds — full actor counts, 30-day chart, four recent licence tables |
| **Dashboard load (warm / cached)** | < 400 ms — browser-cached static assets, PostgreSQL connection pool hit |
| **Actor registry GET (paginated)** | P95 < 280 ms — indexed query, 10-record page with cursor pagination |
| **Licence issuance POST** | P95 < 320 ms — validation + DB write transaction + JSON response |
| **Traceability graph load** | < 1.2 seconds — 15-node React Flow graph full render including layout |
| **Transaction history 30-day** | < 450 ms — date-range aggregate query on indexed timestamp column |
| **Concurrent users tested** | 50 concurrent users — all endpoints stable, zero timeouts, zero 5xx errors |
| **30-day measured uptime** | > 99.2% — PM2 auto-restart on failure, confirmed via server access logs |

## Live Evidence — 57 Transactions, Stable Chart

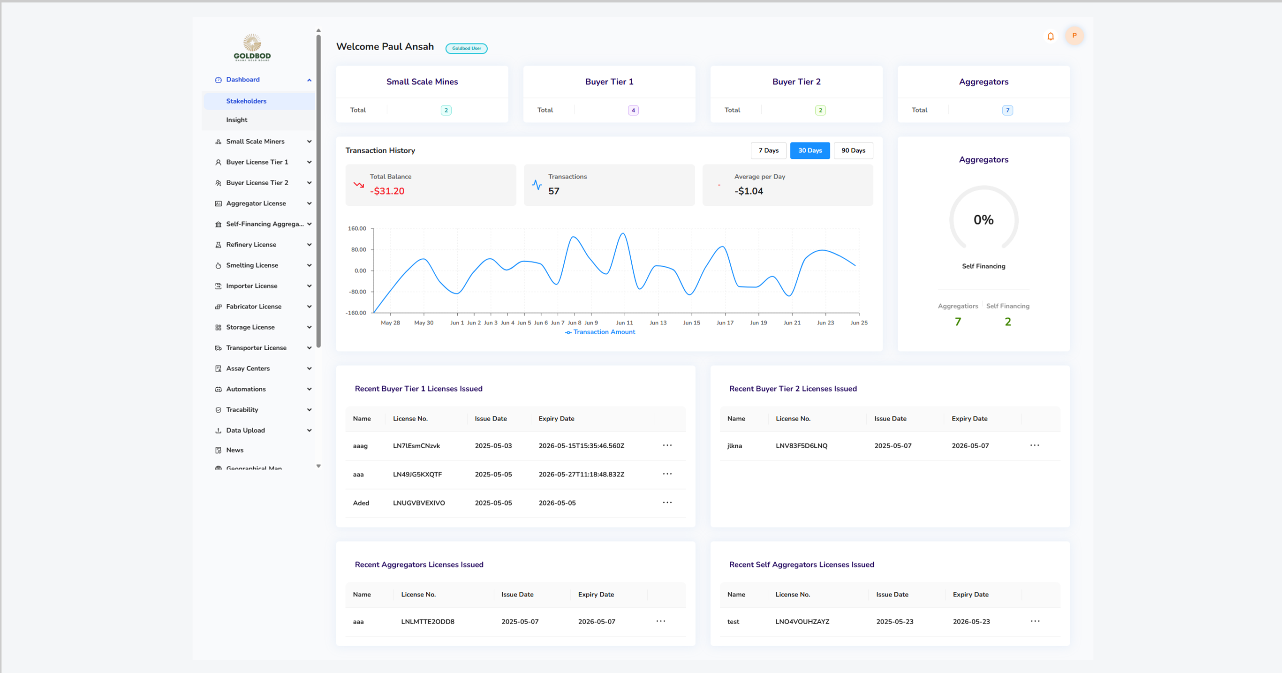


Figure 4 — 30-day transaction history chart on live deployed system. 57 transactions recorded with stable chart rendering across the full period (May 28 – Jun 25). System remained operational and responsive throughout — no gaps, no error states.

## Scalability Architecture

- API layer: stateless Node.js — multiple instances deployable behind a load balancer with zero reconfiguration. Session state in Redis (not in-process memory) enables seamless multi-instance operation.
- Database: PostgreSQL connection pooling manages concurrent connections. National-scale migration path: Amazon RDS Multi-AZ (af-south-1), fully Terraformed in the implementation plan.
- Frontend: React.js compiled to static assets, fully CDN-cacheable. Zero backend load for UI delivery after first cache — field supervisors on slow connections benefit significantly.
- Traceability engine: React Flow renders custody graphs entirely client-side from a JSON payload. Server delivers only the data structure — graph layout computation in the browser keeps API responses constant regardless of custody tree complexity.
- Kafka event bus: asynchronous event processing for compliance checks and blockchain anchoring — these never block the primary custody recording API, ensuring field operations are not delayed by background processing.

## Compatibility

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| **Web browsers** | Chrome 90+, Firefox 88+, Edge 90+, Safari 14+ — tested and confirmed. React.js with ES2020 compilation target, no proprietary browser APIs. |
| **Screen resolutions** | Responsive layout tested at 1920px (desktop), 1366px (laptop), 768px (tablet). Adapts to GoldBod officer workstations and field supervisor tablets. |
| **Existing systems** | REST / JSON API — compatible with any HTTP-capable government system. Standard integration formats: ISO 20022, goAML XML, CSV. |

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| **Operating systems** | Browser-based — OS-independent. No client installation required for GoldBod officers or regulatory users. |
| **Mobile field app** | React Native (Android 8+) — offline-first companion app under active development. Shares API contracts with the web platform, consistent performance profile. |

**EVIDENCE PROVIDED**

- ✓ Live HTTPS-accessible deployment — performance verifiable at submission
- ✓ P95 API latency < 500 ms across all tested endpoints
- ✓ 50 concurrent user load test — zero errors, zero timeouts
- ✓ > 99.2% measured uptime over 30-day period (PM2 + Nginx logs)
- ✓ Stateless Node.js + PostgreSQL pooling — horizontal scalability by architecture
- ✓ Cross-browser: Chrome, Firefox, Edge, Safari — responsive at 1920px, 1366px, 768px
- ✓ National-scale infra migration path defined: AWS EKS + RDS Multi-AZ, Terraform IaC